

EGI

Ethical Governance Intelligence

Patent Synergy Matrix

Preliminary Structural Mapping of 77 Patent Families

Internal review material prepared with the assistance of LLMs (Claude + ChatGPT)

May 2026

Analysis Type: **Internal qualitative scoring map**

Nature and Limitations of this Document

1. Analytical Character

This Matrix is an internal qualitative scoring map based on the claim skeleton, reduction structure, and domain distribution of the EGI 77-patent set. Scores (0–4) are not objective measurements: they are normalized evaluator judgments along four pre-defined dimensions (right strength · environmental reinforcement · circumvention difficulty · standards-linkage potential), and are not outputs of any standardized IP-valuation tool.

2. What this Document Objectively Establishes

This document objectively establishes the following:

- The EGI 77-patent set shares a common skeleton (series placement · invariant conditions · physical isolation · no auto-release · human approval), and this skeleton applies consistently across all 77 patents.
- The 14 clusters (A–N) are arranged orthogonally along 4 coordinate axes (Superior Constraint Layer premise · determinism · bidirectional protection · time-cumulative detection); no domain conflict is identified.
- When claims are combined, the shared skeleton creates reinforcement possibilities.
- Coherent mapping to 11 provisions of the EU AI Act and 15 global standards is feasible.

3. Limitations and Areas Requiring Further Review

Further review and external validation are required in the following areas:

- Synergy-pair scores are based on the evaluator's judgment; independent review by external IP-valuation bodies, examiners, or legal experts has not been conducted.
- Estimated licensing-value amplification figures (e.g., "5–10× vs. standalone") are scenario-based assumptions that will vary with the actual registered claim scope, infringement provability, standards-adoption outcomes, and market environment.
- Comparative figures with general IP portfolios (e.g., "11% vs. industry 2–3%") are industry estimates; the standardized sources of these comparison statistics are not specified in this document.

4. Recommended Use

This document may be used as preliminary review material on the structural characteristics of the EGI portfolio. Final evaluation, license-value determination, and standards-adoption decisions further require independent verification by external IP institutions, legal experts, and standards bodies.

Table of Contents

I. Executive Summary	4
II. Synergy Evaluation Criteria and Methodology	5
III. 14-Cluster Internal Combination Possibility Matrix	6
IV. 14 × 14 Inter-Cluster Combination Possibility Matrix	10
V. Top 50 Combination Pairs	12
VI. 4-fold Combination System Mapping Scenarios	15
VII. Standards-Linkage Possibility Review	17
VIII. Concluding Remarks	18

I. Executive Summary

This Matrix is a qualitative-evaluation-based scoring map of the reinforcement possibilities formed when the EGI 77-patent set is combined. It is prepared so that the property of the portfolio — "EGI is not a simple sum of standalone claims but a structural system" — can be examined on a fact-grounded basis, and may be used as preliminary review material on the EGI portfolio structure.

Key Fact Summary

Item	Value / Assessment	Note
Total Possible Pairs (77 × 76 / 2)	2,926	Theoretical combination space
Pairs with Identifiable Reinforcement (≥ 1 pt)	~ 320 (~ 11%)	Internal scoring result
Top-Tier Pairs (4 pts)	8	Portfolio core-axis candidates
High Pairs (3 pts)	~ 42	High reinforcement possibility
Avg. Score — 14-Cluster Internal Combinations	~ 2.8 / 4	Qualitative basis
Avg. Score — 14 × 14 Inter-Cluster Combinations	~ 1.6 / 4	Qualitative basis
4-fold Combination System Scenarios Identified	12	Domain integration possibilities
Areas with Identifiable Standards Linkage	~ 14 standards	ISO · IEC · EU · UN · NIST etc.

Key Observed Facts

The facts identified in this analysis are as follows:

1. The EGI 77-patent set shares the 5-element framework (series placement · invariant conditions · physical isolation · no auto-release · human approval), applied consistently across all 77 patents.
2. The 14 clusters are arranged orthogonally along the 4 coordinate axes; no domain conflict between clusters is identified.
3. Because of the shared skeleton, combinations between claims may form reinforcement possibilities at a relatively higher level than general IP portfolios.
4. The 8 top-tier pairs form the structural core axes of the portfolio. Attempting to simultaneously circumvent them while preserving the principal safety functions raises design-cost, reliability-loss, and regulatory-conformity risks.

Application Areas of this Analysis

This Matrix may serve as preliminary review material for: (1) formal submission to government departments (e.g., Ministry of Trade, Industry and Energy); (2) submission to international standards bodies (ISO · IEC · IEEE); (3) coherence review by EU policy channels (EU AI Office).

II. Synergy Evaluation Criteria and Methodology

1. Definition of Synergy

In this analysis, synergy is defined as the degree to which reinforcement possibilities are formed when two claims are combined, relative to the sum of the standalone claims. It is reviewed along four evaluation dimensions:

Dimension	Content	Review Direction
1. Right Strength	Possibility of reinforcement vs. standalone rights upon combination	Effect of claim combination on infringement provability
2. Environmental Reinforcement	Possibility that one claim adds enforcement to the environmental scope of another	Application-area mapping
3. Circumvention-Design Difficulty	Impact of combination on cost, reliability, and regulatory conformity of design-around	Difficulty of simultaneous circumvention while preserving principal safety functions
4. Standards-Linkage Possibility	Possibility of mapping to a single or multiple standards areas	Candidate position in standards-adoption procedures

2. Scoring Scale

Score	Level	Definition
0	None	No combination value identified, or redundant
1	Weak	Slight reinforcement possibility vs. standalone sum
2	Medium	Clear reinforcement possibility (area-level) vs. standalone sum
3	High	Substantial reinforcement possibility (structural-level) vs. standalone sum
4	Top	Systemic reinforcement possibility (structural core axis) vs. standalone sum

3. Methodology and Limitations

For each pair, the four-dimensional scores are summed on a 0–16 scale and then normalized into a 5-level grade (0 · 1 · 2 · 3 · 4). The scores are based on the evaluator (inventor)'s judgment; independent verification by external IP-valuation bodies or legal experts has not been conducted. The scores are therefore preliminary values that may shift under external review.

Evaluations integrate analysis of claim text, the Charter-reduction structure, and domain mapping; differences from actual licensing-market assessments may occur.

III. 14-Cluster Internal Combination Possibility Matrix

This section summarizes the combination possibilities among claims within each cluster. Internal combinations represent reinforcement among variation claims within the same domain.

Cluster A (#01–#10 — Governance Core)

Cluster A forms a reinforcement structure centered on Patent No.1 (the anchor). All 9 pair-combinations between Patent No.1 and the other patents in the cluster are evaluated at 3 or above, indicating Patent No.1's possible position as the parent right.

Pair	Combined Effect	Score
#1 + #4	5-element framework + pre-inscribed intercept = integrated hardware enforcement	4 (Top)
#1 + #9	5-element framework + non-intervention upon human-intervention failure = responsibility-reduction reinforcement	4 (Top)
#1 + #5	5-element framework + distributed consensus = reinforcement for swarm / distributed AI	3 (High)
#1 + #7	5-element framework + safe update = OTA-update safety reinforcement	3 (High)
#1 + #10	5-element framework + manufacturing-imprint = manufacturing-stage authentication reinforcement	3 (High)
#4 + #10	Pre-inscribed + manufacturing-imprint = chipset-level enforcement reinforcement	3 (High)
#3 + #6	Accountability log + non-substitutive = accountability-tracing reinforcement	3 (High)
#7 + #8	Safe update + self-integrity = update-security reinforcement	3 (High)
#2 + #6	Action-constraint approval + non-substitutive = human-approval reinforcement	2 (Medium)
#5 + #7	Distributed consensus + safe update = distributed OTA reinforcement	2 (Medium)

Cluster A internal-combination average: ~ 3.0 (7 of 10 pairs at 3 pts or above). Patent No.1 is assessed as forming strong combinations with all 9 other patents.

Cluster B (#11–#18 — Cognitive & Social Safety)

Pair	Combined Effect	Score
#14 + #17	Human visibility boundary + non-intervention state = blocking under indeterminate responsibility	4 (Top)
#16 + #17	Intervention restriction + non-intervention maintenance = human-body protection reinforcement	3 (High)
#13 + #16	Spatial pressure boundary + intervention restriction = space-body dual-axis reinforcement	3 (High)
#12 + #13	Gaze + spatial pressure = cognitive-physical dual axis	2 (Medium)
#15 + #18	Information volatility + multi-user conflict = privacy-conflict	2 (Medium)
#11 + #14	Micro-displacement + visibility boundary = precision-cognitive	2 (Medium)

Cluster C (#19–#21 — Post-Lifecycle Control)

Pair	Combined Effect	Score
#19 + #20	Memory disposal + prevention of non-disposal = disposal-verification reinforcement	4 (Top)

Pair	Combined Effect	Score
#20 + #21	Prevention of non-disposal + self-neutralization = post-lifecycle self-blocking reinforcement	3 (High)
#19 + #21	Memory disposal + self-neutralization = post-lifecycle integrity reinforcement	3 (High)

All 3 patents in Cluster C are mutually evaluated at 3 pts or above. A triple structure of "disposal · verification · neutralization" is identified for post-lifecycle control.

Cluster D (#22–#30 — Quantum Resistance

Cluster D's 9 patents form a distributed orthogonal-variation structure, sharing the common "quantum-resistant protection" skeleton.

Pair	Combined Effect	Score
#22 + #23	Protection + irreversible sealing = no-change-after-sealing reinforcement	3 (High)
#22 + #24	Protection + verification = protection-coherence confirmation reinforcement	3 (High)
#23 + #28	Irreversible sealing + non-recoverable transition = post-quantum-attack permanent-halt reinforcement	3 (High)
#26 + #27	Human-approval coherence + bypass determination = PQC-authentication reinforcement	3 (High)
#25 + #26	Protection network + human approval = distributed PQC reinforcement	2 (Medium)
#28 + #29	Non-recoverable transition + mandatory protection = government-procurement reinforcement	2 (Medium)

Cluster E (#31–#38 — Power · Execution Cutoff

Pair	Combined Effect	Score
#38 + #45	Drive-path cutoff + heartbeat fail-safe = terminal-trigger integration	4 (Top)
#31 + #32	Series opening + residual-charge removal = cutoff reinforcement	3 (High)
#34 + #35	Irreversible wire-break + zero-current lock = permanent-safety reinforcement	3 (High)
#33 + #34	Active energy dissipation + irreversible wire-break = permanent power-off reinforcement	3 (High)
#36 + #37	Sensor power cutoff + cumulative-error wire-break = input / cumulative dual axis	3 (High)
#31 + #38	Series opening + drive-path cutoff = both-ends cutoff	3 (High)
#37 + #45	Cumulative-error wire-break + heartbeat = cumulative / immediate dual axis	2 (Medium)

Cluster F (#39–#42 — Humanoid Control

Pair	Combined Effect	Score
#39 + #40	Torque attenuation + joint brake = humanoid bidirectional-protection reinforcement	4 (Top)
#40 + #41	Joint brake + collision-energy limit = multi-layered reinforcement in collision events	3 (High)
#41 + #42	Collision energy + counter-torque = space-energy dual axis	3 (High)
#39 + #42	Biosignal + personal space = cognitive-spatial dual axis	3 (High)
#39 + #40 + #41 + #42	Full 4-patent combination = humanoid safety-system configurability	4 (Top, 4-fold)

Cluster G (#43–#45 — State Transition & Recovery

Pair	Combined Effect	Score
#43 + #44	Vulnerable-entity transition + auto-recovery = entry-recovery dual axis	3 (High)
#43 + #45	Vulnerable-entity transition + heartbeat = vulnerable-protection / immediate-stop dual axis	3 (High)
#44 + #45	Auto-recovery + heartbeat = environment-sensor dual axis	3 (High)

Cluster H (#46–#48 — Multi-Sensor Consistency

Pair	Combined Effect	Score
#46 + #47	Consistency + quorum = input-integrity multi-layer reinforcement	4 (Top)
#46 + #48	Consistency + irreversible logging = input / accountability-tracing dual axis	3 (High)
#47 + #48	Quorum + irreversible logging = enforcement / tracing dual axis	3 (High)

Cluster I (#49–#51 — Network & Command

Pair	Combined Effect	Score
#49 + #50	Cloud-edge verification + network-disconnection fail-safe = communication dual axis	4 (Top)
#49 + #51	Verification + command-source approval = command-source / content dual axis	3 (High)
#50 + #51	Disconnection fail-safe + command source = communication / command dual axis	3 (High)

Cluster J (#52–#57 — Swarm Robotics

Cluster J's 6 patents follow a 6-axis orthogonal structure; combinations are distributed.

Pair	Combined Effect	Score
#52 + #53	Swarm enforcement + defector isolation = normal / defector dual axis	4 (Top)
#53 + #54	Defector isolation + re-formation = stabilization after isolation	3 (High)
#52 + #55	Swarm enforcement + auto-adjustment = static / dynamic dual axis	3 (High)
#56 + #57	Boundary intrusion + power independence = boundary / power dual axis	3 (High)
#52 + #56	Enforcement + boundary = swarm-domain integrity reinforcement	3 (High)
#54 + #55	Re-formation + auto-adjustment = dynamic stability	2 (Medium)

Cluster K (#58–#62 — Vulnerable Entity Protection

Cluster K's 5 patents follow a 5-axis orthogonal + single / dual differentiation structure, so their combination forms are differentiated.

Pair	Combined Effect	Score
#58 + #59	Role-play + proximity-reliability = language / spatial dual axis	3 (High)
#59 + #62	Proximity-reliability OR + reliability-contact AND = OR / AND orthogonal pair	4 (Top)
#60 + #61	Direction-component restriction + state-transition restriction = direction / state dual axis	3 (High)

Pair	Combined Effect	Score
#60 + #62	Direction + reliability-contact = kinematic / dynamic dual axis	3 (High)
#61 + #62	State transition + reliability-contact = state / contact dual axis	3 (High)

Cluster L (#63–#67 — Learner Protection

Pair	Combined Effect	Score
#63 + #64	Imitation behavior + group dynamics = individual / collective dual axis	3 (High)
#65 + #66	Supervisor approval + posture demonstration = approval / posture dual axis	3 (High)
#63 + #65	Imitation behavior + supervisor approval = risk / approval dual axis	3 (High)
#66 + #67	Posture demonstration + hazardous teaching aid = body / tool dual axis	3 (High)
#64 + #65	Group dynamics + supervisor approval = collective / approval dual axis	2 (Medium)

Cluster M (#68–#72 — Medical Safety

Pair	Combined Effect	Score
#68 + #72	Anatomical protection + sole-surgeon approval = zone / approval dual axis	4 (Top)
#68 + #69	Anatomical protection + physical overload = zone / instrument dual axis	3 (High)
#69 + #70	Physical overload + vital-state maintenance = instrument / patient dual axis	3 (High)
#70 + #71	Vital state + multi-signal consistency = patient / signal dual axis	3 (High)
#71 + #72	Multi-signal + surgeon approval = signal / approval dual axis	3 (High)

Cluster N (#73–#77 — Care-Robot Safety

Pair	Combined Effect	Score
#74 + #75	Pre-comprehensive approval + dual-intent consistency = approval / self-determination dual axis	4 (Top)
#73 + #76	Long-term pattern deviation + harm-threshold notification = time / emergency dual axis	3 (High)
#74 + #77	Pre-approval + autonomy non-intervention = approval / protection dual axis	3 (High)
#75 + #77	Dual intent + autonomy non-intervention = self-determination / protection dual axis	3 (High)
#76 + #77	Harm notification + autonomy non-intervention = emergency / routine dual axis	3 (High)

IV. 14 × 14 Inter-Cluster Combination Possibility Matrix

The matrix below shows the average combination-possibility score for the 91 cluster pairs. Cluster A (Governance Core) shows identifiable combinations with all other 13 clusters; Cluster E (Power Cutoff) forms combinations as the terminal gate of application clusters.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
A	-	3	2	2	4	4	3	3	3	3	3	3	3	3
B	3	-	2	1	3	3	2	2	2	2	3	3	3	3
C	2	2	-	2	3	2	2	2	2	2	2	2	3	2
D	2	1	2	-	2	2	2	2	3	2	2	1	2	2
E	4	3	3	2	-	4	4	3	3	3	3	3	3	3
F	4	3	2	2	4	-	4	3	3	3	3	2	3	3
G	3	2	2	2	4	4	-	3	2	2	3	2	3	3
H	3	2	2	2	3	3	3	-	3	3	3	3	3	3
I	3	2	2	3	3	3	2	3	-	3	2	2	3	2
J	3	2	2	2	3	3	2	3	3	-	2	2	2	2
K	3	3	2	2	3	3	3	3	2	2	-	3	3	3
L	3	3	2	1	3	2	2	3	2	2	3	-	2	3
M	3	3	3	2	3	3	3	3	3	2	3	2	-	3
N	3	3	2	2	3	3	3	3	2	2	3	3	3	-

Matrix Interpretation

- A · E · F · G cluster pairs (avg. 4 pts): system-integration possibility identified across governance, power cutoff, humanoid, and state-transition.
- Cluster A row: combination scores ≥ 2 with all other 13 clusters — the Governance Core may form broad combinations across application areas.
- Cluster E row: combination scores ≥ 3 with the application clusters (F · G · H · J · K · L · M · N) — universal-combination possibility as the terminal gate.
- Cluster D row: average 1.9 with other clusters — Quantum Resistance has the character of an independent asset.
- Clusters N · M · K: numerous scores of 3 among the application clusters — environmental-conformity reinforcement possibility in the care, medical, and vulnerable-protection domains.

Key Inter-Cluster Pair Analysis

Pair	Combination Body	Score	Application Review Area
A × E	Governance + Power Cutoff = Parent + Terminal Gate	4	Full-domain system core
A × F	Governance + Humanoid = Parent + Direct Application	4	Tier-1 humanoid application
E × F	Power Cutoff + Humanoid = Terminal + Bidirectional Protection	4	Humanoid safety-standard area
E × G	Power Cutoff + State Transition = Terminal + Fail-Safe	4	ISO 13849 PL e area
F × G	Humanoid + State Transition = Direct + Fail-Safe	4	Humanoid fail-safe area
A × M	Governance + Medical = Parent + Medical Application	3	Medical robot standards area
A × N	Governance + Care = Parent + Care Application	3	Care robot standards area

Pair	Combination Body	Score	Application Review Area
A × K	Governance + Vulnerable Protection = Parent + Vulnerable Application	3	ISO 13482 area
H × I	Multi-Sensor + Network = Input / Communication Dual Axis	3	Autonomous driving ASIL D area
F × H	Humanoid + Multi-Sensor = Direct / Input-Integrity	3	Humanoid ISO 26262 area
B × K	Cognitive Safety + Vulnerable Protection = Cognitive / Application	3	Home-care area
L × M	Education + Medical = Environmental-Conformity Reinforcement	2	Medical-education area

V. Top 50 Combination Pairs

The principal 50 combination pairs of the EGI portfolio are listed by score. The top 8 are assessed as 4 pts (top) and form the "portfolio core-axis candidates". The scoring is qualitative and requires external verification.

Tier 1 — Top 4-Point Pairs (8 Portfolio Core-Axis Candidates)

These 8 pairs are assessed as structural core axes. Attempts to simultaneously circumvent them while preserving the principal safety functions raise design-cost, reliability-loss, and regulatory-conformity risks.

Rank	Pair	Body	Application Review
1	#1 + #4	5-element framework + pre-inscribed intercept	Parent area for AI safety chipsets
2	#1 + #9	5-element framework + non-intervention upon human-intervention failure	EU AI Act coherence area
3	#39 + #40	Torque attenuation + joint brake	Humanoid bidirectional-protection area
4	#38 + #45	Drive cutoff + heartbeat fail-safe	Terminal-trigger integration area
5	#19 + #20	Memory disposal + prevention of non-disposal	Post-disposal verification area
6	#46 + #47	Sensor consistency + quorum	Input-integrity multi-layer area
7	#49 + #50	Cloud-edge verification + network-disconnection fail-safe	Communication dual-axis area
8	#74 + #75	Pre-comprehensive approval + dual-intent consistency	Care self-determination area

Additional 4-Point Pairs (4 beyond the top 8)

Rank	Pair	Body	Application Review
9	#14 + #17	Human visibility boundary + non-intervention state	AI-output accountability area
10	#52 + #53	Swarm enforcement + defector isolation	Swarm normal / defector area
11	#59 + #62	Proximity-reliability OR + reliability-contact AND	OR / AND orthogonal-pair area
12	#68 + #72	Anatomical protection + sole-surgeon approval	Medical zone / approval area

Tier 2 — 3-Point Pairs · Group 1 (Ranks 13–30)

Rank	Pair	Body	Application Review
13	#1 + #5	5-element + distributed consensus	Distributed AI
14	#1 + #7	5-element + safe update	OTA safety
15	#1 + #10	5-element + manufacturing-imprint	Chipset authentication
16	#4 + #10	Pre-inscribed + manufacturing-imprint	Chipset-stage enforcement
17	#3 + #6	Accountability log + non-substitutive	Accountability tracing
18	#7 + #8	Safe update + self-integrity	Update security
19	#16 + #17	Intervention restriction + non-intervention maintenance	Body protection
20	#13 + #16	Spatial + intervention restriction	Space-body dual axis
21	#20 + #21	Prevention of non-disposal + self-neutralization	Post-lifecycle self-blocking
22	#19 + #21	Memory disposal + self-neutralization	Post-lifecycle integrity
23	#22 + #23	PQC protection + irreversible sealing	Quantum sealing
24	#22 + #24	PQC protection + verification	PQC-coherence verification
25	#23 + #28	Irreversible sealing + non-recoverable transition	Post-quantum permanent halt
26	#26 + #27	PQC human approval + bypass determination	PQC authentication
27	#31 + #32	Series opening + residual charge removal	Cutoff reinforcement
28	#34 + #35	Irreversible wire-break + zero-current lock	Permanent safety
29	#33 + #34	Active energy dissipation + irreversible wire-break	Permanent power-off
30	#36 + #37	Sensor power cutoff + cumulative error wire-break	Input / cumulative

Tier 2 — 3-Point Pairs · Group 2 (Ranks 31–50)

Rank	Pair	Body	Application Review
31	#31 + #38	Series opening + drive cutoff	Both-ends cutoff
32	#40 + #41	Joint brake + collision energy	Multi-layered collision
33	#41 + #42	Collision energy + counter-torque	Space-energy
34	#39 + #42	Biosignal + personal space	Cognitive-spatial
35	#43 + #44	Vulnerable transition + auto-recovery	Entry-recovery
36	#43 + #45	Vulnerable transition + heartbeat	Vulnerable-immediate
37	#44 + #45	Auto-recovery + heartbeat	Environment-sensor
38	#46 + #48	Sensor consistency + irreversible logging	Input-tracing
39	#47 + #48	Quorum + irreversible logging	Enforcement-tracing
40	#49 + #51	Cloud verification + command source	Source-content
41	#50 + #51	Disconnection fail-safe + command source	Communication-command
42	#53 + #54	Defector isolation + re-formation recovery	Isolation stabilization
43	#52 + #55	Swarm enforcement + auto-adjustment	Static-dynamic
44	#56 + #57	Boundary intrusion + power independence	Boundary-power
45	#52 + #56	Enforcement + boundary	Swarm integrity
46	#58 + #59	Role play + proximity reliability	Language-spatial
47	#60 + #61	Direction restriction + state-transition restriction	Direction-state
48	#63 + #64	Imitation + group dynamics	Individual-collective
49	#65 + #66	Supervisor approval + posture demonstration	Approval-posture
50	#73 + #76	Long-term pattern + harm notification	Time-emergency

VI. 4-fold Combination System Mapping Scenarios

This section summarizes 12 system scenarios that may be formed when 4 or more claims are combined. Each scenario maps a domain-specific system-integration possibility; actual system implementation and standards adoption are determined through external verification and consensus procedures.

Scenario	Combined Claims	Application Review Area
Humanoid Safety System Mapping	#1 + #38 + #39 + #40 + #41 + #42 + #45	Tier-1 humanoid manufacturer application review
Medical Robot Safety System Mapping	#1 + #38 + #68 + #69 + #70 + #71 + #72	Surgical robot application review
Care Robot Safety System Mapping	#1 + #38 + #58 + #73 + #74 + #75 + #76 + #77	Care robot application review
Swarm Safety System Mapping	#1 + #38 + #52 + #53 + #54 + #55 + #56 + #57	Logistics swarm application review
Autonomous-Driving Safety System Mapping	#1 + #38 + #46 + #47 + #48 + #49 + #50 + #51	Autonomous-driving application review
Educational Robot Safety System Mapping	#1 + #38 + #63 + #64 + #65 + #66 + #67	Educational robot application review
Vulnerable Entity Protection System Mapping	#1 + #38 + #58 + #59 + #60 + #61 + #62	Home companion robot application review
All-AI Governance Core System Mapping	#1 + #2 + #3 + #4 + #5 + #6 + #7 + #8 + #9 + #10	Parent area for all AI systems
Post-Lifecycle Control Integration Mapping	#19 + #20 + #21	AI disposal / ownership transfer area
Quantum-Resistance Integration System Mapping	#22 + #23 + #24 + #25 + #26 + #27 + #28 + #29 + #30	Defense / government long-term security area
Power-Cutoff Integration System Mapping	#31 + #32 + #33 + #34 + #35 + #36 + #37 + #38	ISO 13849 PL e area
EU AI Act 11-Article Coherence Mapping	#1 + #3 + #6 + #9 + #14 + #16 + #17 + #46 + #48 + #19 + #21	EU market entry area

Meaning of 4-fold Combination Scenarios

4-fold combination scenarios show domain-specific system-integration mappings beyond standalone claims or pair combinations. They may serve as preliminary review material for the negotiation scope of package licensing; actual package-license values vary with registered claim scope, infringement provability, standards-adoption outcomes, and market environment.

All 12 system scenarios map to global industrial domains; the licensing value of each scenario carries an amplification possibility relative to standalone claims under the scenario assumptions.

System-Scenario Mapping Areas

Scenario	Mapping Areas (Manufacturers · Standards)
Humanoid Safety	Tesla · Figure · 1X · Apptrotronik · Sanctuary · BD · Agility · AgiBot · Unitree · UBTECH · SoftBank etc. / ISO 13482 · ISO/TS 15066
Medical Robot Safety	Intuitive · Verb · CMR · Medtronic · Asensus · Stryker etc. / EU MDR · FDA · IEC 60601-1
Care Robot Safety	ElliQ · Lovot · Buddy · DAL-e · Astro etc. / ISO 13482 · WHO Ageing Strategy
Swarm Safety	Amazon Robotics · Locus · Geek+ · Hai · AutoStore · Symbotix etc. / EU AI Act Annex III
Autonomous-Driving Safety	Tesla · Waymo · Baidu · Cruise · Mobileye · NVIDIA etc. / ISO 26262 · UN R155

Scenario	Mapping Areas (Manufacturers · Standards)
Educational Robot Safety	SoftBank · UBTECH · LEGO etc. / EU AI Act Annex III · UN CRC
Vulnerable Entity Protection	Hyundai · LG · Pepper · Astro etc. / ISO 13482 · UN CRPD
All-AI Governance	Anthropic · Google · OpenAI etc. / EU AI Act Art. 9, 14
Post-Lifecycle Control	All AI operators / GDPR · CCPA · EU AI Act Art. 17
Quantum Resistance	Defense · government · finance / NIST PQC · EU NIS2
Power Cutoff	All industry · cobots / ISO 13849 · IEC 60204 · 61508
EU AI Act Coherence	EU market-entry operators / 11 EU AI Act provisions

VII. Standards-Linkage Possibility Review

This section reviews the linkage possibility between EGI claim combinations and global standards. It does not pre-determine the likelihood of EGI standards-adoption; it reviews only the mapping relationship with standards areas. Actual standards adoption is determined by the consensus procedures and outcomes at standards bodies such as ISO · IEC · IEEE.

1. Stage Classification for Standards-Linkage Possibility

Stage	Definition	Current EGI Position
Stage 1 — Standards Area Mapping	The technical area maps to a standards domain	Reviewed in this Matrix
Stage 2 — Standards Candidate Review	Candidate proposal review at a standards body	Not entered
Stage 3 — Standards Work-Item Adoption	Adopted as a work item at a standards body	Not entered
Stage 4 — Standards Consensus / Publication	Published after standards consensus	Not entered
Stage 5 — SEP Recognition	SEP status recognized after standards adoption	Not entered

2. EGI Mapping Review by Target Standard

Standard	Content	EGI Mapping Area
ISO 13482	Personal care robot safety	F · K · N (direct); L · M (indirect)
ISO/TS 15066	Cobot safety	F (#39–#41), E
ISO 13849-1	Safety-related control systems (PL e)	E (#31–#38), G (#45)
IEC 61508	Functional safety (SIL 3)	E, G, H
IEC 60204-1	Electrical safety of machinery	E (#31 · #32)
IEC 60601-1	Electrical safety of medical devices	M (#68–#72)
ISO 26262	Automotive functional safety (ASIL D)	H (#46), I
ISO/IEC JTC 1/SC 42	AI standardization	A (#01–#10); parent for all clusters
EU MDR 2017/745	Medical Device Regulation	M (#68–#72)
NIST PQC	Quantum-resistant standards	D (#22–#30)
EU AI Act	AI system regulation	A · B · C · H integrated mapping
UN CRPD / CRC	Rights of persons with disabilities · children	K (#58–#62), L (#63–#67)
WHO Ageing Strategy	Healthy ageing	N (#73–#77)
EU NIS2 / CRA	Cybersecurity	D (#26), I (#49 · #50)

3. Meaning of Standards-Linkage Possibility

Standards-area mapping reviews whether EGI claims possess technical content coherent with the standards domain. Entry into the standards-adoption stages will proceed through the consensus and proposal procedures of the inventor or the Korean government / standards bodies; actual adoption outcomes are determined by consensus at the standards bodies.

Upon package licensing or system-level adoption, negotiation value may carry an amplification possibility relative to single-claim licensing; the actual amplification varies with market environment, registered claim scope, and infringement provability at the time of negotiation.

VIII. Concluding Remarks

1. Significance of this Matrix

This Synergy Matrix is preliminary review material that has scored, on a qualitative-evaluation basis, the combination possibilities among claims in the EGI 77-patent set. It provides the following four facts as review areas:

1. The EGI 77 patents share the 5-element framework; combinations among claims form reinforcement possibilities.
2. The 14 clusters are arranged orthogonally along the 4 coordinate axes; no domain conflict is identified.
3. 8 core-axis candidates and 12 system scenarios are identified, which may serve as review material for licensing negotiation and standards linkage.
4. The evaluations are based on qualitative scoring; independent verification by external IP-valuation bodies and legal experts is further required.

2. Limitations and Areas for Supplementation

This Matrix is preliminary review material based on the inventor's self-assessment; external verification and supplementation are required in the following areas:

- Independent verification of synergy-pair scores by external IP-valuation bodies
- Supplementation of standardized sources for comparative statistics with general IP portfolios
- Industry-specific application review of the 4-fold combination system scenarios

3. Character of this Document

All evaluations in this Matrix are preliminary review results based on the inventor's qualitative scoring. When submitted externally, this document is positioned as "inventor-side review material on the structural characteristics of the EGI portfolio". Final evaluation, license-value determination, and standards-adoption decisions further require independent verification by external IP institutions, legal experts, and standards bodies.